

The relationship between sleep and physical activity across the lifespan

Taren Sanders*¹, Timothy Hartwig², James Conigrave³, Borja del Pozo-Cruz⁴⁻⁶, Philip Parker⁷, Michael Noetel⁸, Devan Antczak⁹, Stuart J. Fairclough¹⁰, Shona Halson²,
DATASET PROVIDERS TBD¹², & Chris Lonsdale¹³

¹ Institute for Positive Psychology and Education, Australian Catholic University, North Sydney, Australia

² School of Behavioural and Health Science, Australian Catholic University, North Sydney, Australia

³ Centre for Alcohol Policy Research, La Trobe University, Bundoora, Australia

⁴ Faculty of Sport Sciences, Universidad Europea de Madrid, Madrid, Spain

⁵ Faculty of Biomedical and Health Sciences, Universidad Europea de Madrid, Madrid, Spain

⁶ Department of Sports Science and Clinical Biomechanics, University of Southern Denmark, Odense, Denmark

⁷ Office of the Deputy Vice-Chancellor (Research and Enterprise), Australian Catholic University, North Sydney, Australia

⁸ School of Psychology, University of Queensland, Brisbane, Australia

⁹ School of Education, University of Wollongong, Wollongong, Australia

¹⁰ Department of Sport and Physical Activity, Edge Hill University, Ormskirk, United Kingdom

¹¹ *TBD*

¹³ Office of the Provost, Australian Catholic University, North Sydney, Australia

Author Note

Enter author note here.

Correspondence concerning this article should be addressed to Taren Sanders*, 33 Berry St, North Sydney, NSW, Australia. E-mail: Taren.Sanders@acu.edu.au

Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Keywords: keywords

Word count: X

The relationship between sleep and physical activity across the lifespan

Introduction

Sleep is an important determinant of population health [1]. The consequences of poor sleep are profound and include increased all-cause mortality [2], cardiovascular disease [3], cancers [4], and poor mental health [5]. Poor sleep is common: over 50% of children and adolescents [6] and 35% of adults [7] report lower sleep duration than recommended. Improving sleep should be a priority for public health policy and more safe and accessible strategies that promote sleep are needed [8]. In adults, being physically active appears to improve both the quantity and quality of sleep [9] and the World Health Organisation recommends physical activity as a method to improve sleep in adults [10]. In children, adolescents, and older adults associations are unclear [8,11,12] and guidelines do not currently support a role for physical activity to improve sleep.

Promoting physical activity may be an effective method to improve sleep across the lifespan but the published literature to date has at least two limitations that prevent firm conclusions to support health guidelines. First, most studies have relied on self-report measures of sleep and physical activity [8,11,12], which could confound associations and limit the precision of analyses designed to identify a dose-response relationship between physical activity and sleep. Second, no study has collected data from a large sample that included participants across the lifespan. Umbrella reviews [8] and meta-analyses [3,4,9,11–28] have provided some suggestive evidence about how physical activity as a means of improving sleep may vary across the lifespan. However, their reliance on aggregated study-level effects limits the quality and scope of data that can be used to analyse the influence of age [29]. Individual participant data (IPD) analysis can overcome these limitations [30].

In this study we harmonised individual participant data (IPD) from activity monitor observations obtained from studies around the world. The objective of this pooled IPD

study was to investigate the relationship between device-measured physical activity and sleep across the lifespan. We examined whether individuals who are more physically active have better sleep, whether age, sex, and socio-cultural factors moderate the association between physical activity and sleep, and whether there is a bidirectional association between physical activity and sleep.

Methods

We pre-registered all methods and analyses on the Open Science Framework (OSF) before data acquisition began (<https://osf.io/ahd96>).

Data sources and data acquisition

We included data from studies, with any design, that used activity monitors and a 24-hour wear time protocol over a minimum of 4 days, including 1 weekend day. Studies needed to have used research-grade accelerometers from which raw accelerations (m/s^2 or mg) could be obtained. Studies could use any wear location for the accelerometer (hip, wrist, or thigh).

We identified potentially relevant studies by examining the reference lists of recent systematic reviews that explored the relationship between sleep and physical activity in children, adolescents, and adults [9,11,12]. We then used forward searching on potentially eligible studies [31]. We additionally performed a literature search to identify accelerometry consortiums and individual studies, searched trial registries (ANZCTR, ICTRP, ISRCTN) and the Open Science Framework for unpublished results, and we asked collaborators to help identify any additional studies.

We contacted chief investigators or consortium leads by email to request access to deidentified individual data. To make the data acquisition manageable, we only invited studies with sample sizes larger than 400 participants. If required by the contributing research group, the contributing investigators or their university signed a data transfer

agreement. All studies included in the pooled data meta-analysis had to have been approved by their local institutional ethical committees, and all participants had to have provided written informed consent. The lead university's Research Ethics Committee granted permission for each study's data to be obtained, and investigators transferred their deidentified data via a secure file-sharing system.

Data harmonisation and outcomes

All activity monitor data were harmonised by reprocessing raw accelerometer files using an open-source GGIR package in R to produce physical activity and sleep outcomes [32,33]. By reprocessing all raw accelerometer data, we were able to generate outcomes that could be compared across studies, devices wear location, and populations [34]. Data were reprocessed by the core research team, or by data contributors themselves using the GGIR package processing script provided by the core research team. Reprocessed individual study datasets were then pooled into a single dataset. A detailed description of the reprocessing method and decisions for specific accelerometer devices is available in an online supplementary file.

For sleep, GGIR produced the following metrics; total sleep time (minutes), sleep efficiency (percentage of time asleep within the sleep period), sleep regularity (regularity of sleep and awake patterns on a day-to-day basis calculated as the sleep regularity index), and timing (time of sleep and awakening within the 24-hour time cycle). For physical activity, we used volume and intensity metrics proposed by Rowlands et al [35]. These measures are 1) average acceleration (measure of physical activity volume) and 2) intensity gradient (measure of intensity distribution where the natural logs of time and intensity are used to produce a straight-line graph with the slope of the graph being the intensity gradient). We used this approach to model the full continuum of physical activity intensity for our main analyses [36] and avoid collapsing physical activity data into population-specific intensity categories which occurs when intensity cut-points are applied

to accelerometer data [37].

For the purposes of translating physical activity data for interpretation and public health recommendations and for time-use based analyses we then determined; 1) the number of minutes spent in acceleration intensity bands in $50mg$ increments [35], 2) the number of minutes spent in light and moderate to vigorous physical activity using age appropriate cut-points applied to the time in intensity bands data, and 3) the minimum acceleration measured for a range of different durations, this is the MX metric where X refers to the duration. For example, $M60$ refers to the minimum acceleration for the most active 60 minutes of the day [36]. We also categorise all awake time into bout lengths for sedentary, light, and moderate to vigorous physical activity bouts using the categories defined by Gába et al [38]. After extracting sleep and physical activity, the remaining time within the 24-hour time cycle was classified as sedentary behaviour. Non-wear time and abnormally high acceleration values were detected using the data processing default setting [39]. These data were classified as missing and imputed using the accelerometer data from the same time interval on the remaining days of the week.

Covariates and moderators

The following individual participant data were obtained when available; age, sex, height, weight, waist circumference, maturational status, socioeconomic status (SES), ethnicity, screen time, sleep medications, and medically diagnosed conditions known to affect sleep. The country, city, time zone, daylight length, and season in which data were collected were also obtained.

Statistical analysis

To reduce the risk of bias in secondary data analysis of IPD, we used a blind analysis approach [40]. We developed and tested models on a 10% holdout subsample of the dataset. When the analysis code was ready, we conducted the final analysis [41].

We checked all data for implausibly low or high measurements which were excluded if they fell outside of $\pm 4SD$ of the mean, as these likely indicate measurement device failures. We assumed missing covariate and moderator data to be missing at random or completely at random and imputed these missing values using multiple imputation by chained equations with the *mice* package in R [42] to provide 50 imputed datasets. Specifically, we used multilevel predictive mean matching imputation clustered by participants. We used Rubin’s rule to pool the results from analyses conducted using each imputed dataset into one single set of results. For all analyses, we examined 95% confidence intervals and used $P < 0.05$ to assess for statistical significance. We used R (version 4.3.2) for all data cleaning and analysis. We report mean and 95% confidence intervals. For categorical variables, percentages are reported. We examined model diagnostics from all models (see supplementary materials).

To examine the relationship between sleep and physical activity (Research Questions 1-2) we fit mixed-effects models with random intercepts for study ID to account for the nesting of participants in studies [43]. We predict each sleep indicator by each physical activity indicator controlling for sex, body mass index (BMI) and region. Quadratic terms were introduced in each model to allow for curvilinear relationships to be modelled. We interacted each predictor with age to reveal how the relationship between physical activity and sleep changes over the lifespan.

Protocol deviations

This paper addresses Hypotheses 1 and 2 from our registered protocol. We will address remaining hypotheses in future research. To streamline our analysis and reduce redundancy, we made some changes to our protocol while working with the holdout data.

Using Study ID as fixed effect. We initially planned to use Study ID as a fixed effect, given that our study selection was not random. But, as we aimed for our findings to generalise to other studies and populations, we instead opted to include random intercepts

for Study ID which can better model generalisable patterns. This approach allows us to account for variability between studies without fully partialing out study effects, thereby better modelling common phenomena. While this change lessened the strength of model intercepts, it did not significantly alter model slopes. The results using Study ID as a fixed effect (i.e., those that align with the protocol) can be found in the supplementary materials, and are qualitatively similar to the main results.

Sleep indicator variable. Our research examines various metrics of sleep: onset, duration, regularity, and efficiency. We initially planned to merge these variables into a single indicator to simplify analysis. However, no validated indicator existed which could be estimated from our measured variables. We decided against creating one, as we could not validate it with our data. Consequently, our models illustrate the relationship between each aspect of sleep and physical activity, instead of describing a straightforward link between overall sleep quality and physical activity.

Log transformation. We aimed to improve model diagnostics using log transformations of skewed variables. Pilot analysis using holdout data revealed that while diagnostics improved on the log scale, model predictions followed extreme exponential patterns near the edge of the data range when back-transformed. Accordingly, we do not use log transformations in the main paper, however these models can be found in supplementary materials.

Patient and public involvement

Patients and members of the public were not involved in the design, analysis or interpretation of this IPD study using secondary data. Results of this study have broad public health interest, particularly policymakers. Findings from this study will be disseminated through institutional websites, press and media platform releases, and tailored messages to public health organisations and governing bodies.

Results

The aggregated data-set describes 220,612 observations of daily physical activity and sleep from 28,823 unique participants. Table 1 shows demographic information for all participants. Observations were not uniform across the days of the week $\chi^2_{(6)} = 80.75$, $p = < .001$. There were fewer observations on Monday ($z = -7.80$) and Sunday ($z = -3.01$); and more on Wednesday ($z = 2.98$). A table of study characteristics can be found in supplementary materials.

The effects of physical activity volume on sleep

We estimated the effects of physical activity on sleep (RQ1) using mixed-effects models. We estimated the effect of physical activity volume on sleep by age, and the results are presented in Table 2 and Figure 1. There was no meaningful relationship between physical activity volume and sleep duration or sleep efficiency. Physical activity volume and sleep onset had a very weak positive association for younger individuals, but a negative association for older individuals, where sleep onset was reduced among those with the highest physical activity. There was a strong positive association between physical activity volume and sleep regularity, which was strongest among older participants. For participants aged 35 years and above, this link weakened among those with a physical activity volume greater than two standard deviations.

The effects of physical activity intensity on sleep

We estimated how physical activity intensity affects sleep across different age groups. We present the results controlling for sex, SES, and BMI, in Table 2 and Figure 2. We found no evidence that physical activity intensity is related to sleep duration, sleep efficiency, or sleep onset. However, we did find evidence for an association between physical activity intensity and improved sleep regularity, which weakened as age increased.

The effects of sleep duration on physical activity

We estimated the effect of sleep duration on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 3. As age increases, both physical activity volume and intensity decrease. We found no evidence for an association between average sleep duration and physical activity volume.

The effects of sleep efficiency on physical activity

We estimated the effect of sleep efficiency on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 4. We did not find a relationship between sleep efficiency and physical activity volume. However, there was a weak linear association between sleep efficiency and more intense physical activity.

The effects of sleep onset on physical activity

We estimated the effect of sleep onset on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 5. There were weak curvilinear relationships where average sleep onset was linked to the highest levels of physical activity volume and intensity. The U-shaped relationship between sleep onset and physical activity volume attenuated for older participants.

The effects of sleep regularity on physical activity

We estimated the effect of sleep regularity on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 6. There was a J-shaped relationship between sleep regularity and physical activity volume. Participants with below-average sleep regularity tended to have average physical activity volume. Increases in regularity above the average were linked to greater physical activity volume. There was a linear relationship between sleep regularity and physical activity intensity which slightly attenuated with age. Greater sleep regularity was associated with greater

physical activity the following day.

Discussion

In this study, we collated and harmonised device-measured physical activity and sleep data from 20 studies around the world, covering participants across the lifespan. We found no evidence that physical activity volume or intensity are related to sleep duration or efficiency. We did, however, observe relationships between the volume of physical activity an individual accumulated, and their subsequent sleep onset and regularity. Sleep was more regular among people who engaged in more intense physical activity. Further, we found no evidence that sleep duration or efficiency were associated with physical activity volume. Those who slept more efficiently engaged in more intense physical activity the following day. Those who fell asleep earlier and at a more regular time also engaged in more intense physical activity, and engaged in more total physical activity. These findings were largely robust to the decisions made during analysis (see supplementary materials for all analyses).

The lack of association between physical activity and sleep duration or efficiency is particularly notable, as a relationship between these variables has been observed in meta-analyses [8]. For example, [9] found small ($0.22 < d < 0.30$) but significant relationships between physical activity and sleep duration and efficiency in adults using experimental studies for both acute and regular physical activity. This discrepancy could be due to the design of the studies, as the majority of studies which have found a relationship were experimental. That is, physical activity was manipulated and it may be that the routine nature of individuals' physical activity makes associations more difficult to detect. Similarly, most of the studies which examined acute changes to physical activity were conducted in laboratory settings. These studies, while useful for understanding the potential mechanisms, are not ecologically valid and may not reflect the relationship between physical activity and sleep in the real world. Laboratory studies also tend to use more sensitive measures of sleep and physical activity, which may generate less statistical

noise and make relationships easier to detect. It could also be that duration of sleep is set more by parameters external to the individual, such as work schedules or parenting duties, than by individual health behaviours.

We found the most consistent evidence for relationships between sleep onset and regularity with physical activity volume and intensity. This relationship appeared to be bi-directional, with sleep onset and regularity influencing physical activity the following day, and physical activity influencing sleep onset and regularity that night. This is consistent with the idea that sleep and physical activity are mutually reinforcing behaviours [8], or that they are habitual and linked to factors such as the value placed on health. Sleep regularity is a stronger predictor of mortality than sleep duration [44], and so the finding that physical activity is associated with sleep regularity may be particularly important.

Interestingly, the effects of physical activity on sleep regularity were stronger in older adults compared to younger individuals. This age-dependent association suggests that promoting physical activity in older adults could be a particularly effective strategy for improving sleep health, which may contribute to better overall health outcomes [2,3]. Alternatively, it could mean that older adults have fewer of the external constraints (e.g., work schedules, parenting duties) that younger adults have, and so are more able to benefit from the effects of physical activity on sleep. Furthermore, the age-related weakening of the positive effects of physical activity volume beyond a certain threshold implies that moderate rather than high levels of physical activity may be sufficient for promoting sleep in older populations. For younger individuals, physical activity was associated with earlier sleep onset, whereas for older participants, physical activity was linked to later sleep onset. This suggests a nuanced relationship, where age-specific factors (such as changes in circadian rhythm, or age-varying commitments such as work or school) might mediate how physical activity impacts sleep timing. Future public health recommendations could consider tailoring activity guidelines to different age groups to optimise sleep benefits.

These findings have other important implications for public health recommendations. Our results support encouraging physical activity for maintaining regular sleep patterns, especially in older adults. However, the data suggest that focusing solely on increasing physical activity volume or intensity may not necessarily improve other aspects of sleep quality, such as duration or efficiency. Public health messages should, therefore, emphasise the role of physical activity in promoting stable sleep schedules rather than framing physical activity as a universal solution for improving sleep. We note, however, that we did not examine the effects of physical activity on sleep in individuals with sleep disorders or other health conditions. Physical activity interventions may still be a useful treatment for these conditions.

Our study has several strengths, including the use of device-based measures, which reduce the bias associated with self-reported sleep and activity data, and the harmonisation of individual data from multiple studies, conducted in geographically diverse regions and among multiple age groups, allowing for robust and generalisable findings. However, the study is not without limitations. Despite our efforts to harmonise data, variability in accelerometer wear locations and slight differences in data collection protocols across studies may have introduced measurement inconsistencies. Additionally, although we controlled for several potential confounders, there may still be unmeasured factors affecting the observed associations. Finally, while our study captured data from across the lifespan, the limited representation of certain age groups, especially adolescents and young adults, restricts the generalisability of our conclusions to these populations. The limited research on those aged 18-35 is particularly concerning given the high prevalence of poor sleep in this age group [7].

Conclusion

Our findings contribute to the growing body of literature that highlights the complex relationship between physical activity and sleep. While physical activity appears to benefit

sleep regularity, its effects on other sleep metrics are less clear, particularly when considering age differences. Future research should aim to explore mechanisms underlying these associations and identify specific intervention strategies that can effectively enhance both sleep and physical activity behaviours for diverse populations.

References

1. Luyster FS, Strollo PJ, Zee PC, Walsh JK. Sleep: A Health Imperative. *Sleep*. 2012 Jun;35(6):727–34.
2. Huang BH, Duncan MJ, Cistulli PA, Nassar N, Hamer M, Stamatakis E. Sleep and physical activity in relation to all-cause, cardiovascular disease and cancer mortality risk. *British Journal of Sports Medicine*. 2022 Jul;56(13):718–24.
3. Kwok CS, Kontopantelis E, Kuligowski G, Gray M, Muhyaldeen A, Gale CP, Peat GM, Cleator J, Chew-Graham C, Loke YK, Mamas MA. Self-Reported Sleep Duration and Quality and Cardiovascular Disease and Mortality: A Dose-Response Meta-Analysis. *Journal of the American Heart Association*. 2018 Aug;7(15).
4. Ma QQ, Yao Q, Lin L, Chen GC, Yu JB. Sleep duration and total cancer mortality: A meta-analysis of prospective studies. *Sleep Medicine*. 2016 Nov;27–28:39–44.
5. Wakefield JRH, Bowe M, Kellezi B, Butcher A, Groeger JA. Longitudinal associations between family identification, loneliness, depression, and sleep quality. *British Journal of Health Psychology*. 2020 Feb;25(1):1–16.
6. Wheaton AG, Jones SE, Cooper AC, Croft JB. Short Sleep Duration Among Middle School and High School Students - United States, 2015. *MMWR Morbidity and mortality weekly report*. 2018 Jan;67(3):85–90.
7. Liu Y, Wheaton AG, Chapman DP, Cunningham TJ, Lu H, Croft JB. Prevalence of Healthy Sleep Duration among Adults—United States, 2014. *MMWR Morbidity and mortality weekly report*. 2016 Feb;65(6):137–41.

8. Kline CE, Hillman CH, Bloodgood Sheppard B, Tennant B, Conroy DE, Macko RF, Marquez DX, Petruzzello SJ, Powell KE, Erickson KI. Physical activity and sleep: An updated umbrella review of the 2018 Physical Activity Guidelines Advisory Committee report. *Sleep Medicine Reviews*. 2021 Aug;58:101489.
9. Kredlow MA, Capozzoli MC, Hearon BA, Calkins AW, Otto MW. The effects of physical activity on sleep: A meta-analytic review. *Journal of Behavioral Medicine*. 2015 Jun;38(3):427–49.
10. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, Carty C, Chaput JP, Chastin S, Chou R, Dempsey PC, DiPietro L, Ekelund U, Firth J, Friedenreich CM, Garcia L, Gichu M, Jago R, Katzmarzyk PT, Lambert E, Leitzmann M, Milton K, Ortega FB, Ranasinghe C, Stamatakis E, Tiedemann A, Troiano RP, Van Der Ploeg HP, Wari V, Willumsen JF. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *British Journal of Sports Medicine*. 2020 Dec;54(24):1451–62.
11. Antczak D, Lonsdale C, Lee J, Hilland T, Duncan MJ, del Pozo Cruz B, Hulteen RM, Parker PD, Sanders T. Physical activity and sleep are inconsistently related in healthy children: A systematic review and meta-analysis. *Sleep Medicine Reviews*. 2020 Jun;51:101278.
12. Lang C, Kalak N, Brand S, Holsboer-Trachsler E, Pühse U, Gerber M. The relationship between physical activity and sleep from mid adolescence to early adulthood. A systematic review of methodological approaches and meta-analysis. *Sleep Medicine Reviews*. 2016 Aug;28:32–45.

13. Aiello KD, Caughey WG, Nelluri B, Sharma A, Mookadam F, Mookadam M. Effect of exercise training on sleep apnea: A systematic review and meta-analysis. *Respiratory Medicine*. 2016;116:85–92.
14. Banno M, Harada Y, Taniguchi M, Tobita R, Tsujimoto H, Tsujimoto Y, Kataoka Y, Noda A. Exercise can improve sleep quality: A systematic review and meta-analysis. *PeerJ*. 2018;2018(7).
15. Bartel KA, Gradisar M, Williamson P. Protective and risk factors for adolescent sleep: A meta-analytic review. *Sleep Medicine Reviews*. 2015 Jun;21:72–85.
16. Edwards BA, Bristow C, O’Driscoll DM, Wong A-M, Ghazi L, Davidson ZE, Young A, Truby H, Haines TP, Hamilton GS. Assessing the impact of diet, exercise and the combination of the two as a treatment for OSA: A systematic review and meta-analysis. *Respirology*. 2019;24(8):740–51.
17. Gao Y-N, Wu Y-C, Lin S-Y, Chang JZ-C, Tu Y-K. Short-term efficacy of minimally invasive treatments for adult obstructive sleep apnea: A systematic review and network meta-analysis of randomized controlled trials. *Journal of the Formosan Medical Association*. 2019;118(4):750–65.
18. Iftikhar IH, Bittencourt L, Youngstedt SD, Ayas N, Cistulli P, Schwab R, Durkin MW, Magalang UJ. Comparative efficacy of CPAP, MADs, exercise-training, and dietary weight loss for sleep apnea: A network meta-analysis. *Sleep Medicine*. 2017;30:7–14.

19. Iftikhar IH, Kline CE, Youngstedt SD. Effects of exercise training on sleep apnea: A meta-analysis. *Lung*. 2014;192(1):175–84.
20. Janssen X, Martin A, Hughes AR, Hill CM, Kotronoulas G, Hesketh KR. Associations of screen time, sedentary time and physical activity with sleep in under 5s: A systematic review and meta-analysis. *Sleep Medicine Reviews*. 2020 Feb;49:101226.
21. Lederman O, Ward PB, Firth J, Maloney C, Carney R, Vancampfort D, Stubbs B, Kalucy M, Rosenbaum S. Does exercise improve sleep quality in individuals with mental illness? A systematic review and meta-analysis. *Journal of Psychiatric Research*. 2019;109:96–106.
22. Lins-Filho OL, Pedrosa RP, Gomes JML, Dantas Moraes SL, Vasconcelos BCE, Lemos CAA, Pellizzer EP. Effect of exercise training on subjective parameters in patients with obstructive sleep apnea: A systematic review and meta-analysis. *Sleep Medicine*. 2020;69:1–7.
23. Mendelson M, Bailly S, Marillier M, Flore P, Borel JC, Vivodtzev I, Doutreleau S, Verges S, Tamisier R, Pépin J-L. Obstructive sleep apnea syndrome, objectively measured physical activity and exercise training interventions: A systematic review and meta-analysis. *Frontiers in Neurology*. 2018;9(FEB).
24. Rubio-Arias JÁ, Marín-Cascales E, Ramos-Campo DJ, Hernandez AV, Pérez-López FR. Effect of exercise on sleep quality and insomnia in middle-aged women: A systematic review and meta-analysis of randomized controlled trials. *Maturitas*. 2017;100:49–56.

25. Stutz J, Eiholzer R, Spengler CM. Effects of Evening Exercise on Sleep in Healthy Participants: A Systematic Review and Meta-Analysis. *Sports Medicine*. 2019;49(2):269–87.
26. Yang S-Y, Lan S-J, Yen Y-Y, Hsieh Y-P, Kung P-T, Lan S-H. Effects of Exercise on Sleep Quality in Pregnant Women: A Systematic Review and Meta-analysis of Randomized Controlled Trials. *Asian Nursing Research*. 2020;14(1):1–10.
27. Yang P-Y, Ho K-H, Chen H-C, Chien M-Y. Exercise training improves sleep quality in middle-aged and older adults with sleep problems: A systematic review. *Journal of Physiotherapy*. 2012;58(3):157–63.
28. Yang Y, Shin JC, Li D, An R. Sedentary Behavior and Sleep Problems: A Systematic Review and Meta-Analysis. *International Journal of Behavioral Medicine*. 2017;24(4):481–92.
29. Riley RD, Dias S, Donegan S, Tierney JF, Stewart LA, Efthimiou O, Phillippo DM. Using individual participant data to improve network meta-analysis projects. *BMJ Evidence-Based Medicine*. 2023 Jun;28(3):197–203.
30. Ioannidis J. Next-generation systematic reviews: Prospective meta-analysis, individual-level data, networks and umbrella reviews. *British Journal of Sports Medicine*. 2017 Oct;51(20):1456–8.
31. Hinde S, Spackman E. Bidirectional Citation Searching to Completion: An Exploration of Literature Searching Methods. *Pharmacoeconomics*. 2015 Jan;33(1):5–11.

32. Migueles JH, Rowlands AV, Huber F, Sabia S, Van Hees VT. GGIR: A Research Community–Driven Open Source R Package for Generating Physical Activity and Sleep Outcomes From Multi-Day Raw Accelerometer Data. *Journal for the Measurement of Physical Behaviour*. 2019 Sep;2(3):188–96.
33. Van Hees VT, Sabia S, Jones SE, Wood AR, Anderson KN, Kivimäki M, Frayling TM, Pack AI, Bucan M, Trenell MI, Mazzotti DR, Gehrman PR, Singh-Manoux BA, Weedon MN. Estimating sleep parameters using an accelerometer without sleep diary. *Scientific Reports*. 2018 Aug;8(1).
34. Rowlands AV, Plekhanova T, Yates T, Mirkes EM, Davies M, Khunti K, Edwardson CL. Providing a Basis for Harmonization of Accelerometer-Assessed Physical Activity Outcomes Across Epidemiological Datasets. *Journal for the Measurement of Physical Behaviour*. 2019 Sep;2(3):131–42.
35. Rowlands AV. Moving Forward With Accelerometer-Assessed Physical Activity: Two Strategies to Ensure Meaningful, Interpretable, and Comparable Measures. *Pediatric Exercise Science*. 2018 Nov;30(4):450–6.
36. Rowlands AV, Dawkins NP, Maylor B, Edwardson CL, Fairclough SJ, Davies MJ, Harrington DM, Khunti K, Yates T. Enhancing the value of accelerometer-assessed physical activity: Meaningful visual comparisons of data-driven translational accelerometer metrics. *Sports Medicine - Open*. 2019 Dec;5(1).

37. Rowlands AV, Sherar LB, Fairclough SJ, Yates T, Edwardson CL, Harrington DM, Davies MJ, Munir F, Khunti K, Stiles VH. A data-driven, meaningful, easy to interpret, standardised accelerometer outcome variable for global surveillance. *Journal of Science and Medicine in Sport*. 2019 Oct;22(10):1132–8.
38. Gába A, Dygrýn J, Štefelová N, Rubín L, Hron K, Jakubec L, Pedišić Ž. How do short sleepers use extra waking hours? A compositional analysis of 24-h time-use patterns among children and adolescents. *International Journal of Behavioral Nutrition and Physical Activity*. 2020 Dec;17(1).
39. Van Hees VT, Gorzelniak L, Dean León EC, Eder M, Pias M, Taherian S, Ekelund U, Renström F, Franks PW, Horsch A, Brage S. Separating Movement and Gravity Components in an Acceleration Signal and Implications for the Assessment of Human Daily Physical Activity. Müller M, editor. *PLoS ONE*. 2013 Apr;8(4):e61691.
40. Weston SJ, Ritchie SJ, Rohrer JM, Przybylski AK. Recommendations for Increasing the Transparency of Analysis of Preexisting Data Sets. *Advances in Methods and Practices in Psychological Science*. 2019 Sep;2(3):214–27.
41. MacCoun R, Perlmutter S. Blind analysis: Hide results to seek the truth. *Nature*. 2015 Oct;526(7572):187–9.
42. Buuren SV, Groothuis-Oudshoorn K. **Mice** : Multivariate Imputation by Chained Equations in *R*. *Journal of Statistical Software*. 2011;45(3).
43. Curran PJ, Hussong AM. Integrative data analysis: The simultaneous analysis of multiple data sets. *Psychological Methods*. 2009;14(2):81–100.

44. Windred DP, Burns AC, Lane JM, Saxena R, Rutter MK, Cain SW, Phillips AJK. Sleep regularity is a stronger predictor of mortality risk than sleep duration: A prospective cohort study. SLEEP. 2024 Jan;47(1):zsad253.

Table 1

Participant characteristics

Characteristic	Age group				
	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	9,758	3,062	473	4,675	6,563
Age	9.49 (1.51)	13.71 (2.39)	23.90 (6.13)	56.72 (7.55)	72.21 (4.73)
BMI	17.95 (3.30)	20.39 (4.00)	24.71 (5.88)	26.87 (4.95)	27.08 (4.42)
Valid wear time hours	22.18 (2.35)	22.18 (2.67)	22.15 (2.63)	22.75 (2.23)	23.82 (0.75)
PA intensity	-2.12 (0.19)	-2.20 (0.19)	-2.37 (0.19)	-2.49 (0.20)	-2.67 (0.22)
PA volume	43.68 (25.61)	42.87 (16.63)	40.64 (11.15)	39.56 (13.10)	31.27 (10.38)
Sleep duration	446.73 (92.90)	403.02 (96.97)	365.62 (79.72)	400.68 (63.47)	398.74 (71.46)
Sleep efficiency	0.79 (0.10)	0.83 (0.10)	0.87 (0.07)	0.88 (0.06)	0.86 (0.06)
Sleep onset	20.96 (1.17)	22.30 (1.96)	24.00 (1.47)	23.65 (1.35)	24.41 (1.30)
Sleep regularity	54.54 (12.67)	54.12 (13.05)	54.75 (12.83)	59.41 (11.52)	54.73 (12.55)
Accelerometer Wear Location					
Wrist	2,479 (25.40%)	1,928 (62.97%)	473 (100.00%)	4,675 (100.00%)	6,563 (100.00%)
Hip	7,279 (74.60%)	1,134 (37.03%)	-	-	-
Ethnicity					
Non-white	1,830 (18.75%)	469 (15.32%)	34 (7.19%)	87 (1.86%)	263 (4.01%)
Unclear	4,016 (41.16%)	1,620 (52.91%)	235 (49.68%)	1,317 (28.17%)	2,475 (37.71%)
White	3,912 (40.09%)	973 (31.78%)	204 (43.13%)	3,271 (69.97%)	3,825 (58.28%)
Region					
Europe	3,203 (32.82%)	1,189 (38.83%)	234 (49.47%)	4,305 (92.09%)	6,560 (99.95%)
Oceania	1,761 (18.05%)	552 (18.03%)	123 (26.00%)	370 (7.91%)	3 (0.05%)
South america	2,074 (21.25%)	984 (32.14%)	116 (24.52%)	-	-

Table 1 continued

Characteristic	Age group				
	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Africa	876 (8.98%)	167 (5.45%)	-	-	-
Asia	610 (6.25%)	90 (2.94%)	-	-	-
North america	1,234 (12.65%)	80 (2.61%)	-	-	-
Season					
Autumn	3,183 (32.62%)	663 (21.65%)	54 (11.42%)	1,149 (24.58%)	1,404 (21.39%)
Spring	2,263 (23.19%)	1,599 (52.22%)	264 (55.81%)	1,156 (24.73%)	2,128 (32.42%)
Summer	1,223 (12.53%)	314 (10.25%)	86 (18.18%)	1,175 (25.13%)	1,358 (20.69%)
Winter	3,089 (31.66%)	486 (15.87%)	69 (14.59%)	1,195 (25.56%)	1,673 (25.49%)
Sex					
Female	5,118 (52.45%)	1,593 (52.02%)	306 (64.69%)	2,619 (56.02%)	2,725 (41.52%)
Male	4,640 (47.55%)	1,469 (47.98%)	167 (35.31%)	2,056 (43.98%)	3,838 (58.48%)
Sleep Conditions Reported					
Yes	49 (0.50%)	1 (0.03%)	-	261 (5.58%)	682 (10.39%)
Socioeconomic Status					
Low	3,607 (36.96%)	819 (26.75%)	65 (13.74%)	1,162 (24.86%)	2,490 (37.94%)
Medium	2,670 (27.36%)	1,105 (36.09%)	265 (56.03%)	2,221 (47.51%)	2,569 (39.14%)
High	3,481 (35.67%)	1,138 (37.17%)	143 (30.23%)	1,292 (27.64%)	1,504 (22.92%)

Note. 88 participants had missing age data. For categorical variables the value is the count, and percentage. For

numeric variables the value is the Mean and SD. N = 28,823

Table 2

Physical activity predicting sleep controlling for SES, sex, and BMI.

Term	Physical activity volume (z)			Physical activity intensity (z)				
	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	0.24 [0.06, 0.41]	0.09	2.69	.007	0.24 [0.07, 0.40]	0.08	2.85	.004
Physical activity	0.10 [0.08, 0.12]	0.01	10.88	< .001	0.08 [0.07, 0.10]	0.01	10.50	< .001
Age	0.00 [0.00, 0.00]	0.00	-3.00	.003	0.00 [0.00, 0.00]	0.00	-2.92	.004
Physical activity ²	-0.01 [-0.01, 0.00]	0.00	-4.94	< .001	0.01 [0.00, 0.02]	0.00	2.76	.006
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-3.57	< .001	0.00 [0.00, 0.00]	0.00	-5.73	< .001
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	-0.05	.959	0.00 [0.00, 0.00]	0.00	-4.89	< .001
Sleep efficiency (z)								
(Intercept)	0.26 [0.09, 0.44]	0.09	2.92	.004	0.29 [0.10, 0.47]	0.10	2.99	.003
Physical activity	0.10 [0.09, 0.12]	0.01	11.30	< .001	0.06 [0.05, 0.08]	0.01	7.97	< .001
Age	0.00 [0.00, 0.00]	0.00	-0.18	.855	0.00 [0.00, 0.00]	0.00	-0.86	.391
Physical activity ²	-0.02 [-0.02, -0.02]	0.00	-11.97	< .001	-0.03 [-0.04, -0.02]	0.00	-6.69	< .001
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-10.55	< .001	0.00 [0.00, 0.00]	0.00	-7.17	< .001
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	4.79	< .001	0.00 [0.00, 0.00]	0.00	5.40	< .001
Sleep onset (z)								
(Intercept)	0.00 [-0.27, 0.27]	0.14	0.01	.995	0.00 [-0.26, 0.26]	0.13	0.01	.988
Physical activity	-0.03 [-0.04, -0.01]	0.01	-3.61	< .001	-0.06 [-0.07, -0.05]	0.01	-10.30	< .001
Age	0.00 [0.00, 0.00]	0.00	-1.73	.083	0.00 [0.00, 0.00]	0.00	-1.22	.222
Physical activity ²	-0.01 [-0.01, 0.00]	0.00	-4.30	< .001	-0.03 [-0.04, -0.02]	0.00	-9.25	< .001
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-11.48	< .001	0.00 [0.00, 0.00]	0.00	1.35	.178
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	8.83	< .001	0.00 [0.00, 0.00]	0.00	11.29	< .001
Sleep regularity (z)								

Table 2 continued

Term	Physical activity volume (z)				Physical activity intensity (z)			
	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	0.53 [0.38, 0.67]	0.07	7.18	< .001	0.71 [0.53, 0.89]	0.09	7.60	< .001
Physical activity	0.24 [0.22, 0.25]	0.01	25.42	< .001	0.27 [0.25, 0.29]	0.01	29.15	< .001
Age	0.00 [0.00, 0.00]	0.00	1.06	.293	0.00 [-0.01, 0.00]	0.00	-3.08	.003
Physical activity ²	-0.03 [-0.03, -0.03]	0.00	-15.45	< .001	-0.10 [-0.11, -0.09]	0.01	-18.83	< .001
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-1.48	.141	0.00 [0.00, 0.00]	0.00	-21.58	< .001
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	-5.89	< .001	0.00 [0.00, 0.00]	0.00	9.47	< .001

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the column headers.

Table 3 continued

Term	Physical activity volume (z)			Physical activity intensity (z)				
	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	1.59 [1.25, 1.94]	0.18	9.08	< .001	1.45 [1.31, 1.58]	0.07	20.69	< .001
Sleep regularity	0.11 [0.09, 0.12]	0.01	12.20	< .001	0.08 [0.07, 0.09]	0.01	13.75	< .001
Age	-0.02 [-0.03, -0.02]	0.00	-38.26	< .001	-0.03 [-0.03, -0.03]	0.00	-48.34	< .001
Sleep regularity ²	0.00 [-0.01, 0.00]	0.00	-0.43	.668	0.00 [-0.01, 0.00]	0.00	-2.11	.035
Sleep regularity $\times$ age	0.00 [0.00, 0.00]	0.00	-10.39	< .001	0.00 [0.00, 0.00]	0.00	-12.13	< .001
Age $\times$ Sleep regularity ²	0.00 [0.00, 0.00]	0.00	-0.05	.959	0.00 [0.00, 0.00]	0.00	0.53	.598

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the row headers.

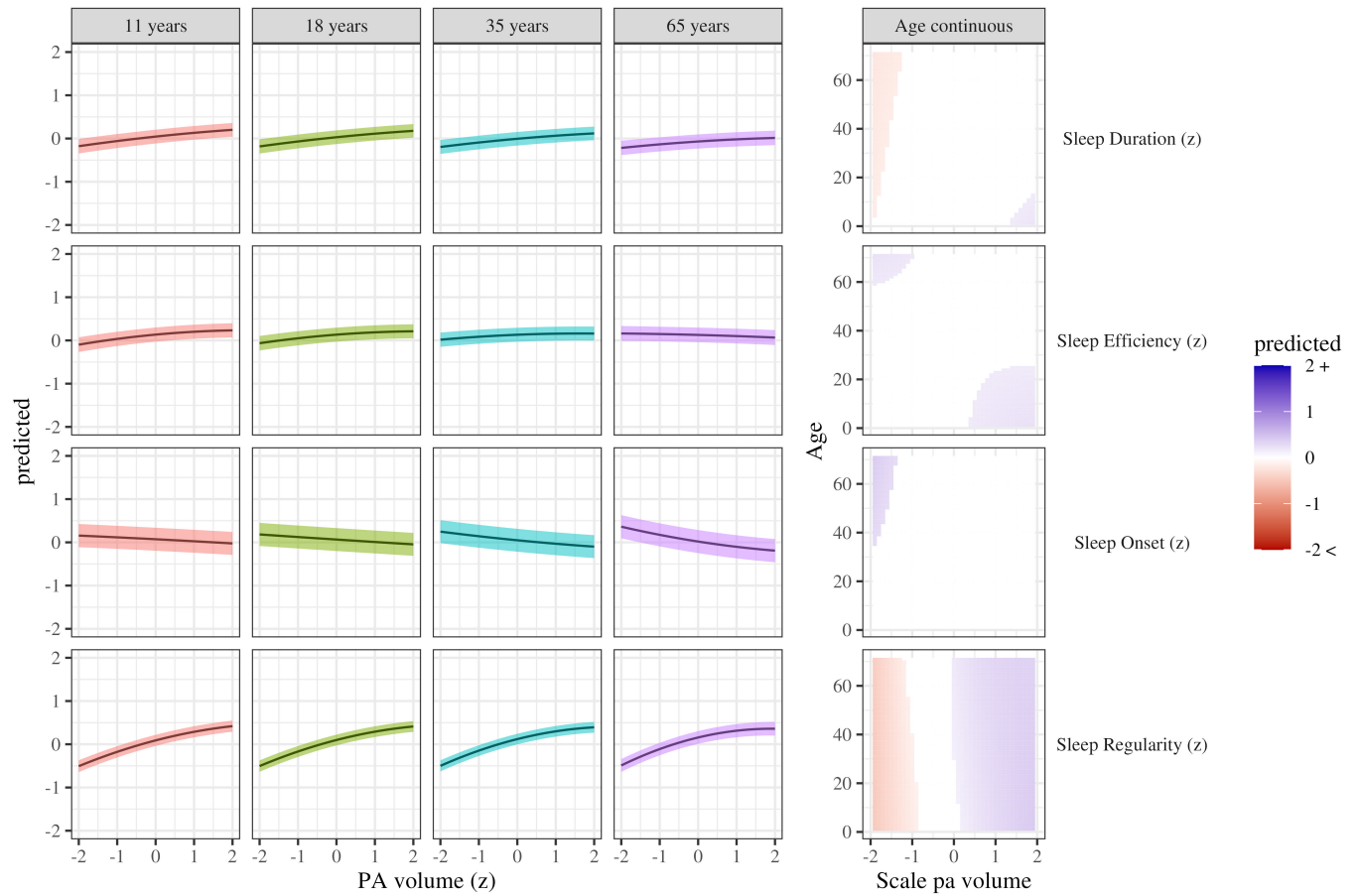


Figure 1. Sleep metrics on physical activity volume. The panels on the left show the curvilinear relationship between PA volume and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero. These plots demonstrate exact turning points where the effects of PA volume change by age.

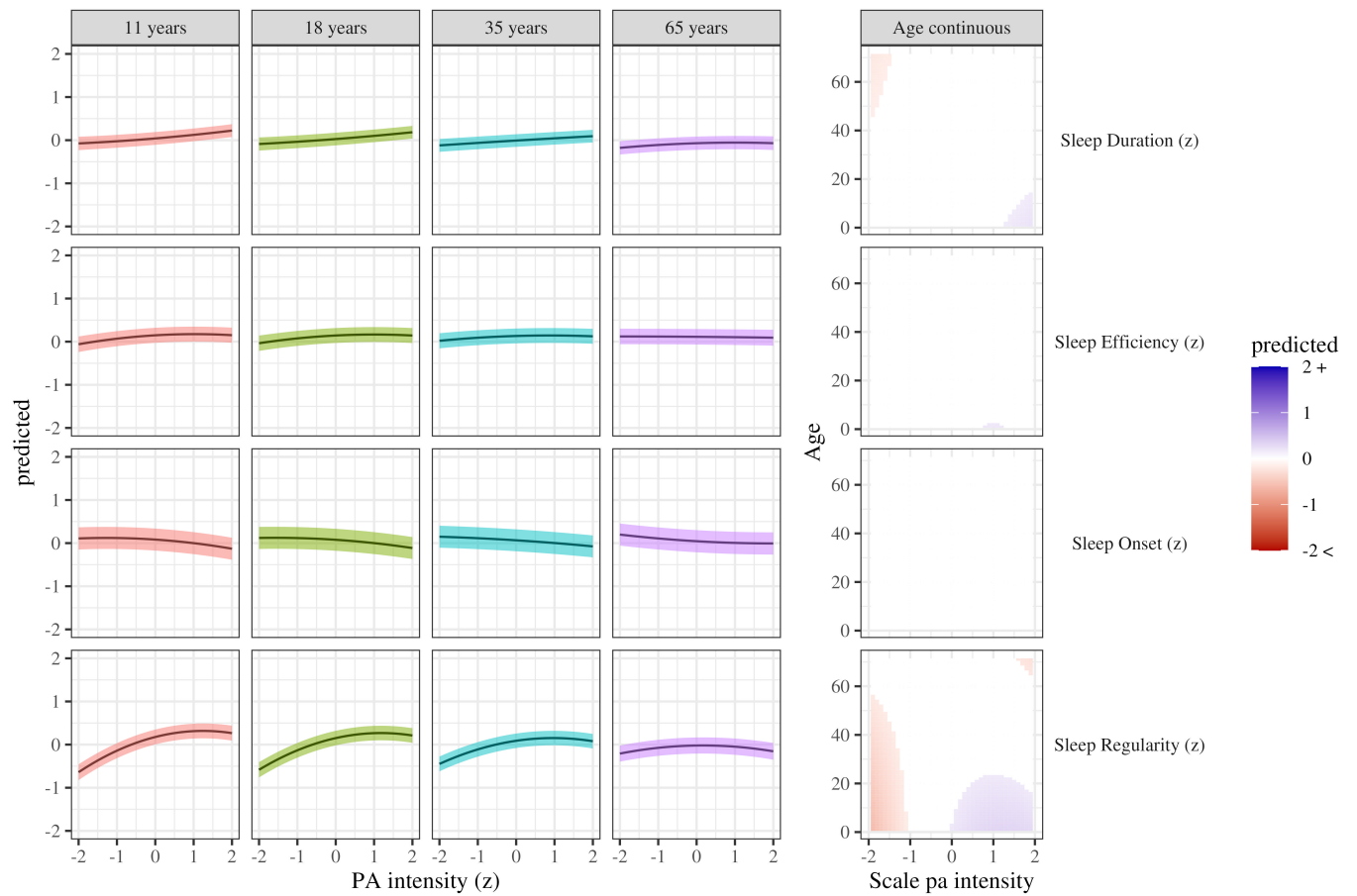


Figure 2. Sleep metrics on physical activity intensity. The panels on the left show the curvilinear relationship between PA intensity and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

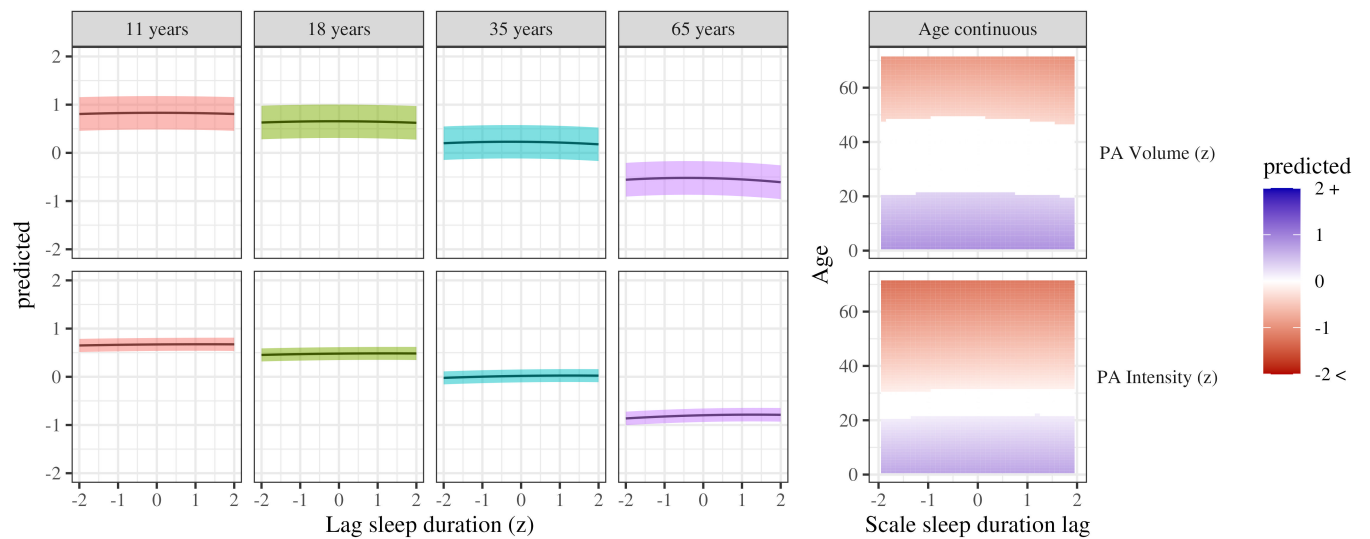


Figure 3. Physical activity by sleep duration. The panels on the left show the curvilinear relationship between sleep duration and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

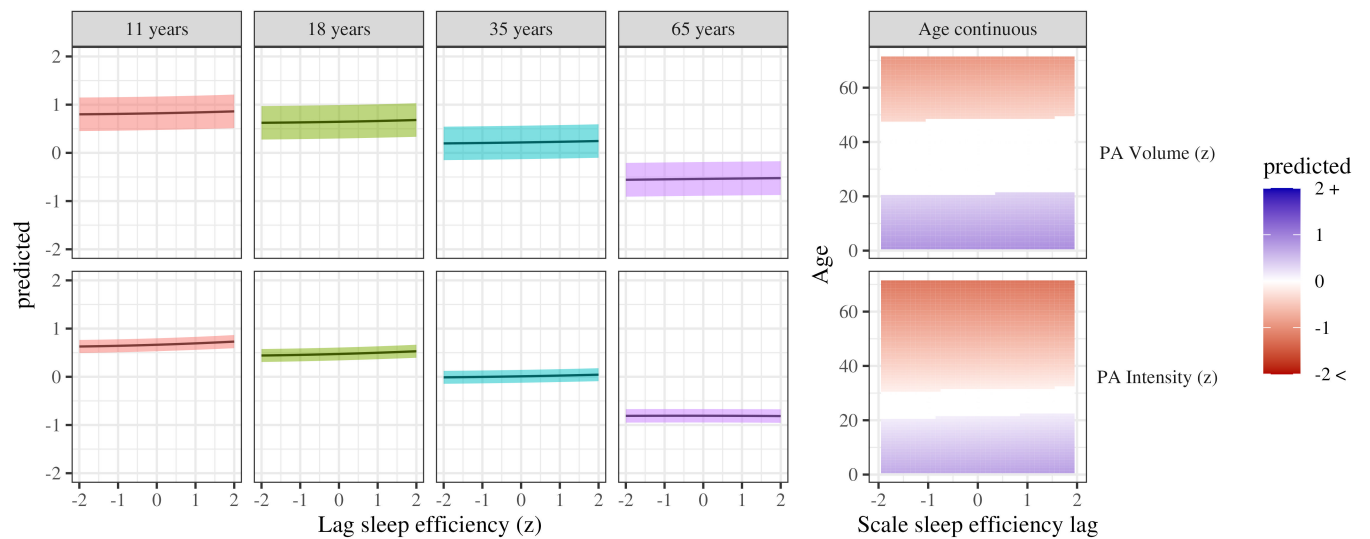


Figure 4. Physical activity by sleep efficiency. The panels on the left show the curvilinear relationship between sleep efficiency and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

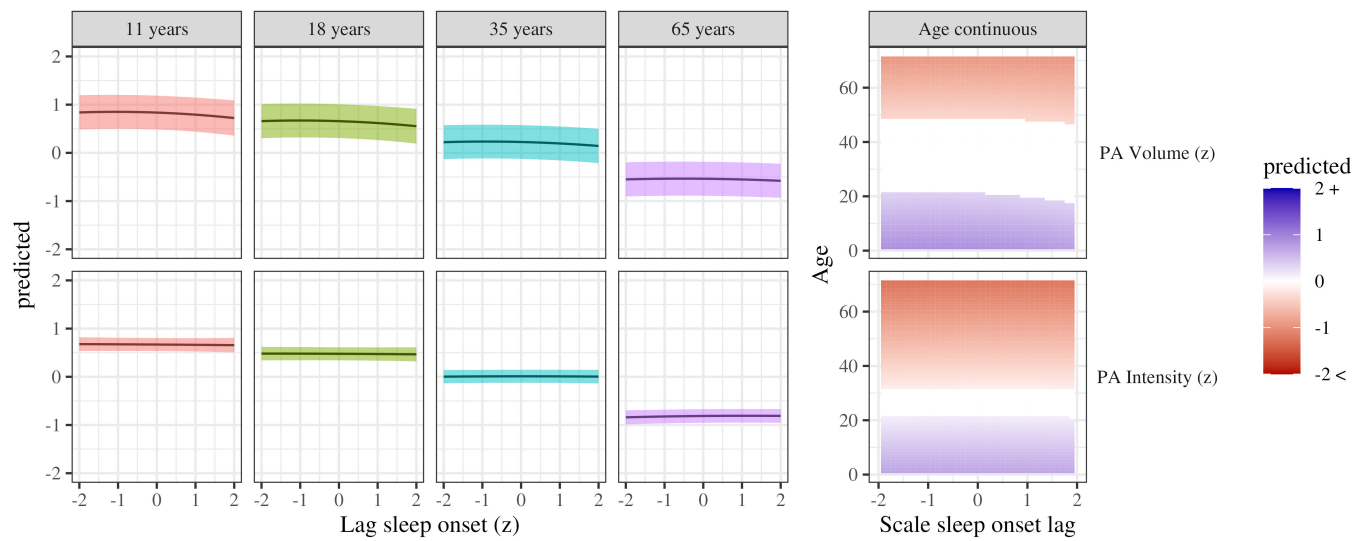


Figure 5. Physical activity by sleep onset. The panels on the left show the curvilinear relationship between sleep onset and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

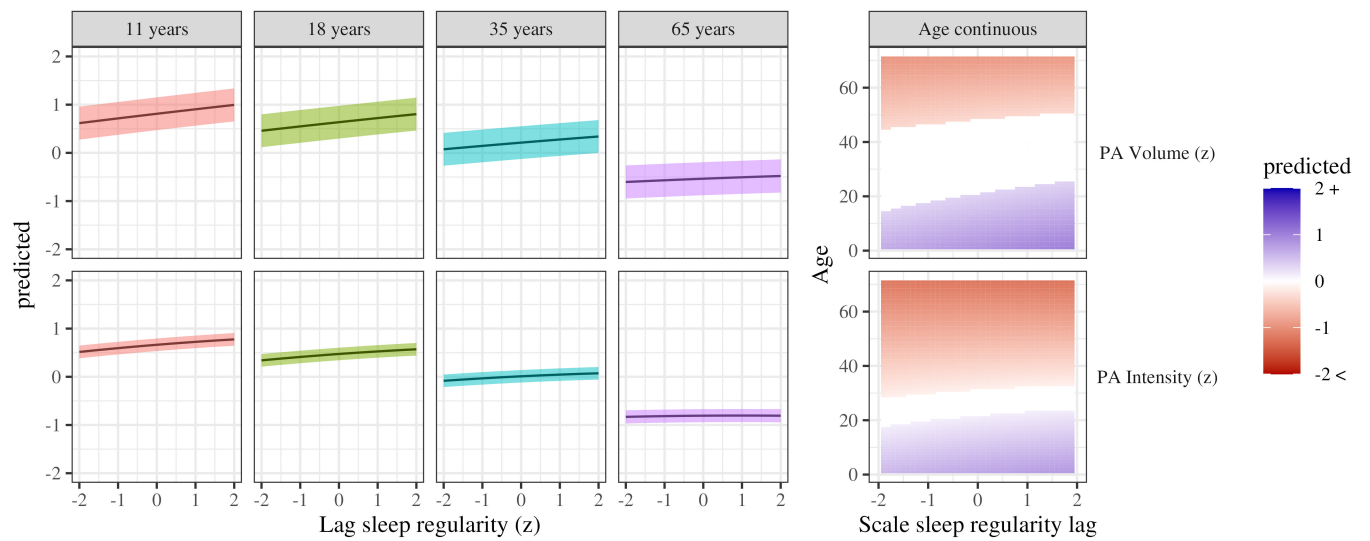


Figure 6. Physical activity by sleep regularity. The panels on the left show the curvilinear relationship between sleep regularity and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.